VAR analysis of impact of 2016 failed coup in Turkey on macroeconomic indicators, civil liberties and institutions

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Programme Evaluation

January 12, 2026

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1 Introduction

1.1 Context and Background

The Turkish economy, a prominent emerging market characterized by its integration into global capital flows and its strategic geopolitical position, experienced an institutional shock on July 15, 2016. This attempted coup marked a turning point toward political uncertainty and increased authoritarianism. At the time of the shock, Turkey already exhibited significant structural vulnerabilities, including a high dependency on external financing and persistent inflationary pressures.

1.2 Motivation of the Study

The nexus between institutional quality and economic performance is well-established in international political economy. However, this research is motivated by the need to disentangle the high-frequency macroeconomic responses from long-term institutional shifts. By utilizing a two frequency approach with annual institutional data with quarterly macroeconomic data. This study aims to isolate the specific transmission channels through which political instability propagates. Understanding whether the 2016 shock acted as a transitory disturbance or a structural break in variables such as industrial production and the balance of payments is essential for quantifying country risk in emerging economies.

1.3 Expected Results and Contributions

Based on Vector Autoregression (VAR) modeling, several key findings are anticipated. Institutionally, the study expects to document a significant and durable degradation in civil liberties and freedom of expression following the shock. Socially, a sharp increase in repression metrics, specifically incarceration rates, is hypothesized as a direct consequence of the post-coup state of emergency. Macroeconomically, Granger causality tests suggest that the shock primarily destabilized the balance of payments, which in turn served as the lead predictor for the contraction in real industrial production. These results contribute to the literature by providing empirical evidence of the ”institutional-to-economic” spillover effect in a major emerging power.

1.4 Organization of the Thesis

The remainder of this paper is organized as follows. Section 2 reviews the theoretical and empirical literature on political shocks and economic performance. Section 3 describes the datasets, covering annual indicators (2010–2022) and quarterly series (2010–2020). Section 4 details the econometric specification of the VAR models, stationarity tests, and the identification of structural shocks. Sections 5 and 6 present the descriptive statistics and the core Impulse Response Function (IRF) analysis. Finally, Section 8 discusses the policy implications for emerging market stability, followed by concluding remarks in Section 7.

2 Literature Review

2.1 The Multidimensional Aftermath of the 2016 Failed Coup

The failed coup attempt of July 15, 2016, represents a break in Turkey's modern history, triggering a transformation that goes beyond political instability. It simultaneously impacted the country's macroeconomic indicators, institutional integrity, and the human capital.

2.1.1 The Economic Importance: Uncertainty and Macroeconomic Volatility

From a macroeconomic perspective, the significance of the 2016 event is rooted in the "Uncertainty Channel" popularized by Bloom (2009). In the Turkish context, Bakirtas and Akpolat (2024) use ARDL bounds testing to show that coup attempts have a permanent negative effect on long-run growth. The importance of studying 2016 specifically lies in the immediate "confidence shock" it triggered.

My research analyzes how institutional shocks trigger capital flight and a 'sudden stop' in financial flows, which subsequently destabilize the Balance of Payments (BOP) and contract Industrial Production. In the context of Turkey's post-2016 transition toward more authoritarianism, to compensate for the loss of foreign direct investment (FDI) and the rising risk premium, the state resorted to credit-driven stimuli (Credit Guarantee Fund), which temporarily sustained growth at the cost of chronic inflation (CPI) and institutional erosion of the Central Bank's independence.

2.1.2 Institutional and Political Transformation: The Security State

The subject is crucial for political economy because it allowed for a total restructuring of the state. Yağcı (2017) develops a "local theory" for Turkey, explaining that the failed coup facilitated "coup-proofing" strategies. This led to the 2017 constitutional referendum and the transition to an executive presidency, which centralizes decision-making and destabilize the climate necessary for investment.

Furthermore, Bulut (2020) identify a "spectacular shock doctrine" used by the state. Through the capture of media and the orchestration of public confessions, the government reengineered social affect to legitimize the dismantling of administrative meritocracy. This institutional decay is not merely political; it is an economic drain that reduces the efficiency of the bureaucracy and the rule of law.

2.2 Sociological and Psychological Dimensions: The Destruction of Human Capital

A unique and vital contribution of recent literature is the focus on the KHK (Decree Laws) and their role in "social exclusion." The dismissal of over 130,000 public servants is not just a human rights issue but a massive destruction of human capital.

- **Civil Death and Stigma:** Avincan (2023) and Fidan (2025) document how the "Us vs. Them" polarization leads to the "civil death" of victims. The stigma of being labeled a "terrorist" excludes highly educated professionals (professors, judges, engineers) from the labor market, leading to a loss of Total Factor Productivity (TFP).

- **The Brain Drain:** Girdap (2020) emphasizes the ”dislocation” of intellectuals. The forced migration of Turkey’s elite to Europe and the US represents a permanent transfer of wealth and knowledge away from the Turkish economy.
- **Psychological and Existential Impact:** Stollwerck (2024) provide evidence of the severe psychological trauma, including torture and enforced disappearances Avincan (2024). The resulting ”social trauma” breaks the trust required for economic transactions and increases the social cost of the state’s survivalist ideology.

2.3 Summary of Key Academic Contributions

The following table summarizes the core literature used to evaluate the impact of the 2016 failed coup across different dimensions.

Table 1: Summary of Literature: Theories, Methodologies, and Key Findings

Author(s)	**Title**	**Methodology**	**Main Findings**
Londregan & Poole	Poverty, the Coup Trap, and Seizure of State Power	Quantitative Analysis (Pooled Time-Series)	Identified a "coup trap" where low per-capita income increases coup probability, which in turn hinders economic development .
Alesina & Perotti	Income Distribution, Political Instability, and Investment	Cross-country Econometric Analysis (2SLS)	Income inequality fuels social unrest, which reduces investment and growth by increasing political instability.
Metin-Özcan et al.	Coup d'État and Economic Growth in Turkey	Time-series Econometrics (VAR / Dummy variables)	Political shocks in Turkey significantly reduced short-term growth and shifted long-term structural parameters .
Acemoglu et al.	A Political Economy of Coups d'État	Theoretical Game Theory (Formal Modeling)	Coups are used by elites as a defensive mechanism to prevent wealth redistribution in nascent democracies .
Waldner & Lust	Unwelcome Change: Democratic Backsliding	Theoretical Literature Synthesis	Democratic backsliding is often an incremental, state-led process of dismantling institutions rather than a sudden rupture.
Opotow	Moral Exclusion and Injustice	Qualitative Psychological Theory	"Othering" removes marginalized groups from the boundary of justice, enabling systemic rights violations without public outcry.
AST Reports	Various (KHK, Stigma, and Human Rights)	Case Studies and Survey Analysis	Systematic "social death" of purged individuals (KHK victims) leads to severe health decline and increased suicidal ideation .

2.4 Contribution of the Analysis

While the existing literature extensively documents the political aftermath of the 2016 failed coup (Stollwerck, 2024) and the general macroeconomic volatility in Turkey (Yağcı, 2017), this study provides a novel contribution by bridging the gap between political science and international macroeconomics through a multi-dimensional econometric approach.

First, this research moves beyond traditional GDP-centric analyses by integrating institutional and sociologival variables such as civil liberties, incarceration rates, and subjective hope levels—into a formal Vector Autoregression (VAR) framework. This allows for a quantitative assessment of social and institutional concepts ,often relegated to qualitative studies (Girdap, 2020).

Second, from a methodological standpoint, the study employs a dual-scale temporal analysis. By utilizing an annual VAR for structural institutional shifts and a quarterly VAR for macroeconomic transmission channels, I identify the specific role of the Balance of Payments (BOP) as a primary conduit of political shocks to the real economy.

Finally, this work contributes to the coup d'état literature by providing empirical evidence of how a failed coup can trigger a break in institutional quality and economic indicators.

3 Data

In this research, I employ a dataset covering the period 2010–2022 to evaluate the impact of the 2016 political shock. I use quarterly data for inflation, production volume and Balance of Payment and annual data for others variables I didn't found quarterly.

3.1 Data Transformation

To ensure the internal validity of the Vector Autoregression (VAR) framework and the comparability of heterogeneous variables, I performed a three-stage data transformation process.

3.1.1 Standardization

Given that the dataset encompasses variables with fundamentally different units—ranging from index points for institutional quality to monetary values for tourism revenue and rates per 100,000 inhabitants for incarceration—I transformed all raw series into standardized Z-scores. This procedure is defined as:

$$std_x_t = \frac{x_t - \mu_x}{\sigma_x} \tag{1}$$

where μ_x and σ_x represent the sample mean and standard deviation, respectively. This standardization allows the Impulse Response Functions (IRFs) to be interpreted in terms of standard deviation shocks, providing a uniform scale for analyzing the magnitude of the 2016 coup's impact across different sectors.

3.1.2 Stationarity and Differencing

A core requirement for a stable VAR system is that all endogenous variables must be stationary ($I(0)$). Upon conducting Augmented Dickey-Fuller (ADF) tests, I found that macroeconomic variables such as Industrial Production and the CPI exhibited non-stationary trends in levels. To address this, I applied first-differencing ($d_$) to the quarterly series:

$$d_std_x_t = std_x_t - std_x_{t-1} \tag{2}$$

This transformation effectively shifts the focus of the analysis from absolute levels to quarterly growth rates and variations, ensuring that the structural shocks are not confounded by long-term deterministic trends.

3.2 Summary Table

The following table summarizes the variables, their frequencies, and their respective sources:

Table 2: Summary of Variables and Data Sources

Variable	**Frequency**	**Source**
CPI (Inflation)	Quarterly	FRED
Production Volume	Quarterly	FRED
Balance of Payment	Quarterly	FRED
FDI Inflows	Annual	World Bank
Tourism Revenue	Annual	World Bank
Crime Rate	Annual	Macrotrends
Incarceration Rate	Annual	World Prison Brief
Corruption Index	Annual	Our World in Data
Civil Liberties	Annual	Our World in Data
Freedom of Expression	Annual	Our World in Data
Hope Level	Annual	TurkStat (TUIK)

4 Methodology

4.1 VAR Model Specification and Rationalization

In this study, I employ a Vector Autoregression (VAR) model to quantify the dynamic interactions between institutional instability and macroeconomic performance in Turkey. I favor this approach because it allows for the endogenous modeling of all variables, capturing the feedback loops where economic deterioration can exacerbate institutional fragility. Unlike traditional structural models, I let the data reveal the dynamic relationships through a system of linear equations, which is crucial in the context of an emerging economy undergoing abrupt shocks.

4.1.1 Mathematical Representation

I specify my VAR model in its reduced form as follows:

$$Y_t = A_0 + \sum_{i=1}^{p} A_i Y_{t-i} + BD_t + \epsilon_t \tag{3}$$

Where:

- Y_t is my vector of endogenous variables representing the three pillars of the Turkish macroeconomy. For my quarterly analysis, I define $Y_t = [std_d_cpi_t, std_d_prod_t, std_d_bop_t]'$, where each variable is standardized and first-differenced to ensure stationarity.
- A_0 is the vector of constants (intercepts).
- A_i represents the matrices of coefficients that I estimate for each lag i, measuring the influence of past values on the current equilibrium of the system.
- D_t is my exogenous dummy variable representing the 2016 attempted coup. I consider this parameter as a pure "institutional innovation," determined outside the economic system.
- B measures the contemporaneous impact of this political shock on my endogenous variables.

- ϵ_t represents the vector of structural disturbances, which I assume to be Gaussian white noise.

4.1.2 The Endogenous Vector and Transmission Channels

I select the variables in the vector Y_t to isolate the "uncertainty channel" triggered by the 2016 shock. The Consumer Price Index (CPI) allows me to observe the monetary response, while industrial production (Prod) serves as a proxy for real activity. Finally, I integrate the Balance of Payments (BOP) as the critical link to global capital markets.

Through this model, I aim to verify three primary transmission mechanisms:

1. **The Investment Channel:** Where political uncertainty prompts firms to adopt a "wait-and-see" approach, leading to a contraction in production.

2. **The Financial Channel:** Where institutional instability triggers capital flight, immediately impacting the dynamics of the BOP.

3. **The Price Channel:** Where the currency depreciation following a financial shock passes through into domestic inflation.

4.2 Lag Selection Criteria

A critical step in VAR modeling is the selection of the optimal lag length p. Selecting too few lags may lead to residual autocorrelation and biased coefficients, while selecting too many lags unnecessarily exhausts the degrees of freedom.

To determine the lag order, I rely on a battery of information criteria, specifically the Akaike Information Criterion (AIC) and the Schwarz-Bayesian Information Criterion (SBIC). While the SBIC tends to favor more parsimonious models, the AIC is often more appropriate for capturing the complex persistence of political shocks. In my quarterly analysis, the tests suggested an optimal lag of 2.

5 Descriptive Statistics

5.1 Overview of Macroeconomic and Institutional Trends

Before proceeding with the Vector Autoregression (VAR) estimation, it is essential to examine the distributional properties and temporal trends of the variables in my dataset. The Turkish economy during the 2010–2022 period is characterized by significant volatility, particularly following the 2016 institutional shock.

As shown in the summary statistics (Table 3), quarterly macroeconomic variables exhibit high variance. Industrial production and the Consumer Price Index (CPI) display persistent upward trends, which justified my decision to use first-differences to achieve stationarity in the econometric model. The Balance of Payments (BOP) data reveals frequent fluctuations, reflecting Turkey's sensitivity to international capital flows and sudden changes in investor sentiment.

5.2 Institutional and Social Divergence

The annual indicators highlight a clear demarcation before and after the 2016 attempted coup. Institutional variables, such as *civil liberties* and *freedom of expression*, show a decline in their mean values in the post-shock period. Similarly, social metrics capture the repressive response to the crisis; notably, the prison rate per 100,000 inhabitants shows a substantial increase, reaching its historical peak in the years following the coup.

Preliminary analysis suggests a correlation between the deterioration of institutional indices and a decline in public optimism, with a shift from the "hopeful" category toward "not hopeful at all" during periods of heightened political tension.

5.3 Comparative Analysis of Pre- and Post-Shock Periods

The descriptive statistics provide initial empirical evidence of a structural break in Turkey's institutional and social landscape following the 2016 attempted coup. By bifurcating the sample into pre-shock ($Post_2016 = 0$) and post-shock ($Post_2016 = 1$) periods, I observe a stark divergence in nearly all indicators.

5.3.1 Institutional Deterioration

The decline in institutional quality is the most prominent feature of the post-2016 era. The mean value for *civil liberties* plummeted from 0.654 to 0.367, while *freedom of expression* experienced an even more severe contraction, dropping from 0.595 to 0.188. Simultaneously, the corruption index rose from 0.627 to 0.841, reflecting a perceived erosion of the rule of law. These shifts are accompanied by a significant reduction in standard deviations (from 0.142 to 0.013 for free speech), suggesting a persistent and systematic institutional tightening rather than transient volatility.

5.3.2 Social and Economic Responses

The social metrics capture the intensity of the judicial response. The mean of prison rate more than doubled, increasing from 147.4 to 323.2 per 100,000 inhabitants, marking a massive expansion of the penal system. On the economic front, FDI inflows declined from a mean of 1.82% to 1.30% of GDP, illustrating the cooling of investor confidence in the wake of heightened political risk. Interestingly, tourism revenue remained relatively resilient in the long term, though it exhibits the highest overall standard deviation ($7,984$), indicating its extreme sensitivity to periodic security shocks.

5.4 The 2016 Multidimensional Shock: A Cross-Sectoral Analysis

To substantiate the choice of 2016 as a major structural break, Figure 1 illustrates the immediate annual variation (%) across all categories in the dataset. This visualization captures the synchronized nature of the institutional, social, and economic disruptions.

Table 3: Descriptive Statistics: Pre- and Post-2016 Comparison

	Pre-2016 (N=13)		**Post-2016 (N=9)**		**Total Sample**	
Variable	**Mean**	**Std. Dev.**	**Mean**	**Std. Dev.**	**Mean**	**Std. Dev.**
Civil Liberties	0.654	0.074	0.367	0.024	0.536	0.155
Free Expression	0.595	0.142	0.188	0.013	0.428	0.231
Corruption Index	0.627	0.097	0.841	0.011	0.715	0.130
FDI Inflows	1.822	0.914	1.299	0.283	1.608	0.760
Prison Rate	147.38	48.41	323.21	47.83	208.93	98.01
Crime Rate	4.298	0.607	2.750	0.377	3.718	0.931
Tourism Revenue	28,744	7,028	30,095	10,676	29,166	7,984
Hope Level	2.692	0.067	2.664	0.080	2.682	0.071

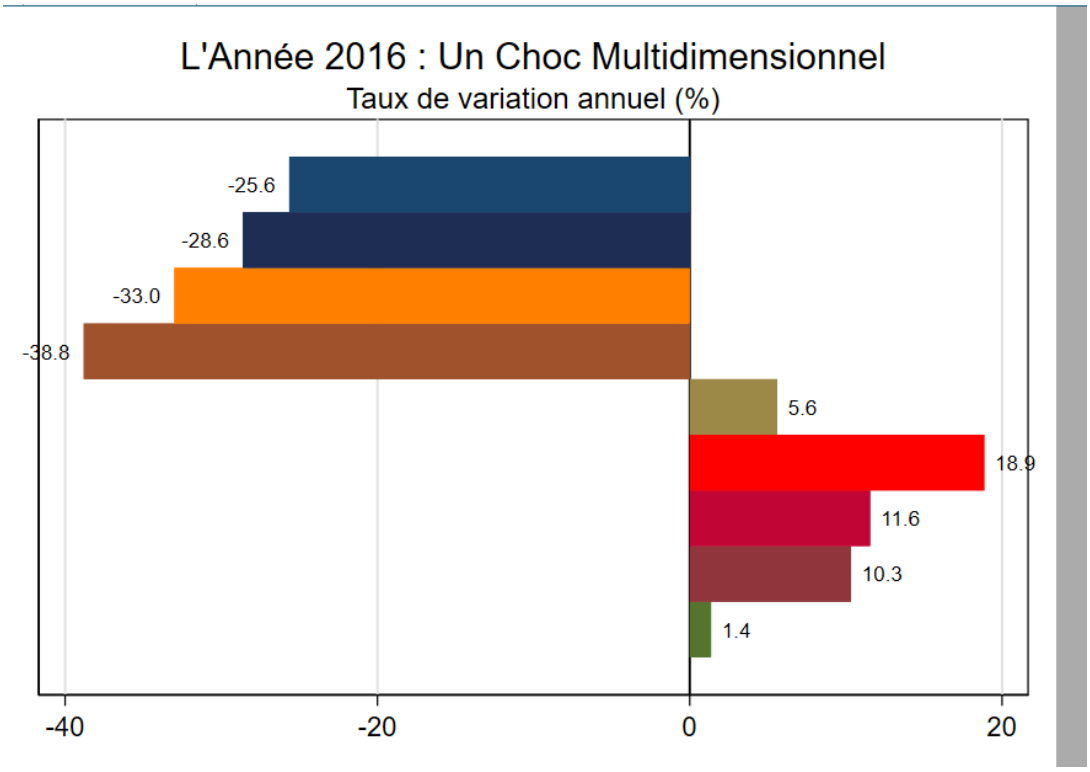


Figure 1: The 2016 Structural Break: Annual Percentage Variation

Line 1: Tourism
Line 2: FDI
Line 3: Civil Liberties
Line 4: Freedom of Expression
Line 5: Corruption
Line 6: Crime Rate (Red)
Line 7: Total Prison Population
Line 8: Incarceration Rate
Line 9: Hope Level

The empirical evidence from this shock year reveals three critical dynamics:

- **Institutional Rupture:** The most pronounced contractions are observed in the institutional block. *Freedom of Expression* experienced the sharpest decline (-38.8%), followed closely by *Civil Liberties* (-33.0%). These figures validate the hypothesis of a sudden institutional tightening in response to the coup attempt.

- **Economic and Capital Outflows:** The economic sector mirrors this instability with a significant contraction in *FDI inflows* (-28.6%) and *Tourism revenue* (-25.6%). This reflects an immediate capital flight and a degradation of Turkey's international attractiveness as investors adopted a "wait-and-see" strategy.

- **The Penal and Security Surge:** Social indicators highlight a surge in *Insecurity metrics* (+18.9% in crime rate) during the crisis year. Furthermore, the expansion of the judicial apparatus is already visible through the double-digit growth in both *Total Prison Population* (+11.6%) and the *Incarceration Rate* (+10.3%), setting the stage for the long-term penal surge identified in the descriptive statistics (Figure 8).

- **Social Sentiment Resilience:** Paradoxically, *Hope Levels* remained positive (+1.4%). This suggests a potential "rally-around-the-flag" effect or a psychological lag in the population's perception of the long-term economic consequences of the shock.

5.5 Social Sentiment and the Evolution of Hope

To capture the psychological dimension of the 2016 shock, I analyze the evolution of "Hope Levels" among the Turkish population. While macroeconomic indicators react instantly to volatility, social sentiment often reflects a combination of immediate political reactions and long-term structural decay.

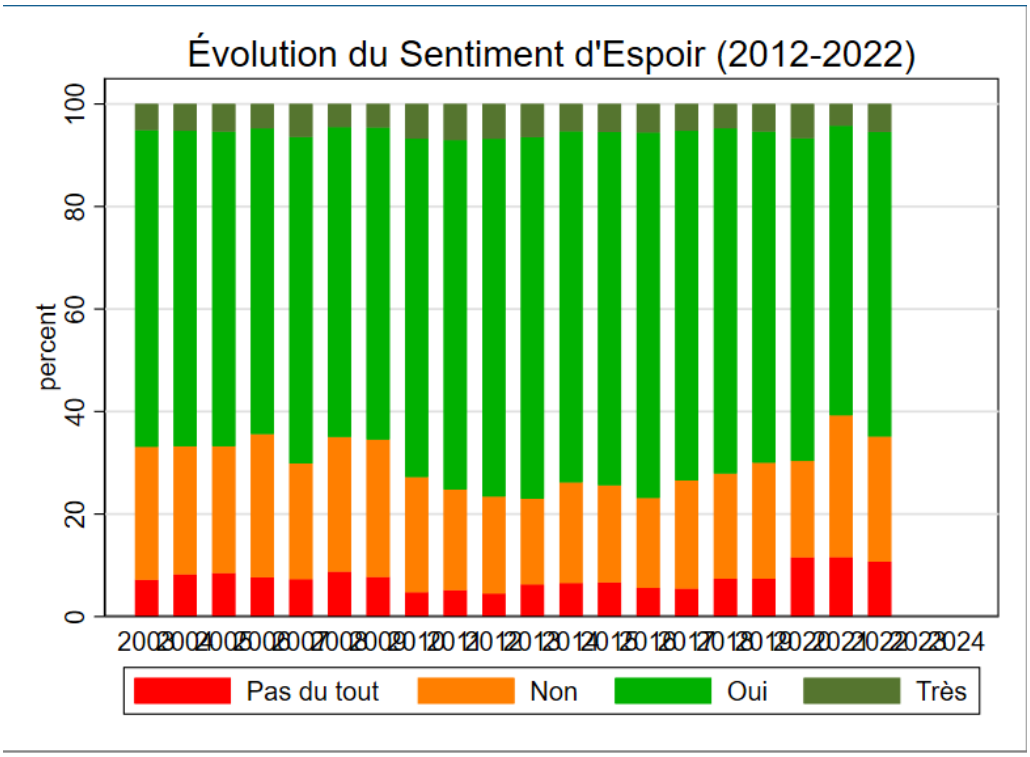


Figure 2: Evolution of Hope Sentiment in Turkey (2012–2022)

The distribution of hope levels, illustrated in Figure 2, reveals several critical phases:

- **Pre-Shock Stability (2012–2015):** The majority of the population remained within the "Hopeful" (Oui) and "Very Hopeful" (Très) categories, despite existing institutional tensions.

- **The 2016 Resilience:** Paradoxically, the year of the attempted coup did not trigger an immediate collapse in hope. The "Not Hopeful" (Non) category saw a slight expansion, but the dominant sentiment remained positive. This aligns with the "rally-around-the-flag" effect where national crises can temporarily consolidate social cohesion behind the state.

- **Long-term Erosion (2019–2022):** A significant shift is observed in the post-2019 period. The "Not Hopeful at All" (Pas du tout) category, represented in red, reaches its historical peak in 2021 and 2022, exceeding 10% of the population. Combined with the "Non" category, negative sentiment now encompasses nearly 40% of the sample.

This trend suggests that while the 2016 political shock was the catalyst, the subsequent erosion of civil liberties and the economic instability (identified in my VAR models) have had a cumulative, lagging negative effect on the Turkish social psyche.

6 Results

6.1 Diagnostic Tests and Model Stability

To ensure the validity of the impulse responses, I first assess the stability of the VAR models. As shown in Figure 6, all eigenvalues of the companion matrix lie within the unit circle, confirming that the systems are stationary. Furthermore, the Breusch-Godfrey LM tests indicate no serial correlation in the residuals ($p > 0.10$), validating the lag specifications for both annual and quarterly models.

6.2 Annual Analysis:

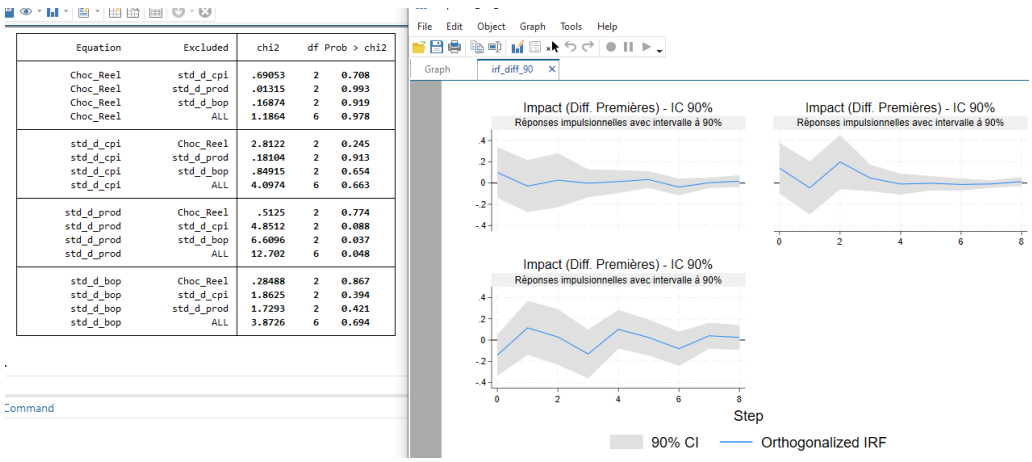
- **Institutional Block:** A one-standard-deviation shock to the 2016 dummy leads to a significant drop in *civil liberties* and *free expression* (Figure 7). The significance of these responses at the 90% confidence level persists for multiple periods, suggesting a permanent shift in Turkey's political regime (Figure 7).

- **Social Block:** The *prison rate* exhibits a significant positive response, nearly doubling in the periods following the shock (Figure 8). The lack of a significant response in the *crime rate* confirms that this surge is a direct consequence of the post-coup judicial measures rather than changes in criminal activity.

- **Annual Economic Block:** FDI and Tourism Dynamics

  The analysis of the annual economic variables provides a nuanced view of Turkey's international standing. Unlike the institutional indicators, the economic variables exhibit higher volatility and wider confidence intervals (Figure 12). **Foreign Direct Investment (FDI):** The IRF for std_fdi shows a negative trajectory following the shock (Figure 12). While the mean inflow decreased from 1.82% to 1.30% of GDP as seen in the descriptive statistics, the annual VAR does not reach statistical significance at the 90% level (Figure 12). This suggests that the impact on FDI is either more gradual or masked by idiosyncratic annual factors not captured in this specification. **Tourism Revenue:** The response of $std_tourism$ is characterized by a temporary contraction followed by a return toward the baseline (Figure 12). The non-significance here emphasizes the "transitory shock" nature of tourism to political events; while 2016 saw a dip, the sector's long-term

resilience prevents a statistically significant permanent shift in the annual model (Figure 3).



Equation	Excluded	chi2	df	Prob > chi2
Choc_Reel	std_d_cpi	.69053	2	0.708
Choc_Reel	std_d_prod	.01315	2	0.993
Choc_Reel	std_d_bop	.16874	2	0.919
Choc_Reel	ALL	1.1864	6	0.978
std_d_cpi	Choc_Reel	2.8122	2	0.245
std_d_cpi	std_d_prod	.18104	2	0.913
std_d_cpi	std_d_bop	.84915	2	0.654
std_d_cpi	ALL	4.0974	6	0.663
std_d_prod	Choc_Reel	.5125	2	0.774
std_d_prod	std_d_cpi	4.8512	2	0.088
std_d_prod	std_d_bop	6.6096	2	0.037
std_d_prod	ALL	12.702	6	0.048
std_d_bop	Choc_Reel	.28488	2	0.867
std_d_bop	std_d_cpi	1.8625	2	0.394
std_d_bop	std_d_prod	1.7293	2	0.421
std_d_bop	ALL	3.8726	6	0.694

Figure 3: VAR results of CPI, Production and BOP

6.3 Quarterly Analysis: The Macroeconomic Transmission

The high-frequency data isolates the financial-to-real transmission of the shock. The Impulse Response Functions (IRF) derived from the quarterly VAR model allow for a dynamic visualization of how the 2016 shock propagates through the Turkish economy over a 10-quarter horizon.

Shock to the Balance of Payments (BOP): The 2016 structural break triggers an immediate and statistically significant contraction in the standardized *Balance of Payments* (std_d_bop). This financial shock persists for approximately four to six quarters before returning to the baseline. This initial reaction confirms the financial risk channel, where political instability leads to capital flight and a degradation of the country's external financial standing.

Impact on Industrial Production : The *Industrial Production* index (std_d_prod) exhibits a negative response with a lag of approximately two quarters. This confirms the Granger-causality results ($p = 0.037$) and suggests that the institutional shock hinders real economic activity through a "credit and uncertainty channel." As external liquidity tightens and country risk rises, industrial investment and output contract, demonstrating the real costs of institutional decay.

Impact on Inflation (CPI): The shock also induces a positive and persistent pressure on the *Consumer Price Index* (std_d_cpi). The shock thus fuels domestic inflation.

6.4 Granger Causality

To investigate the statistical precedence between the variables, we conduct Granger causality Wald tests. This procedure determines whether the past values of one variable provide statistically significant information to forecast another. The results are summarized in Table 4.

First, the results confirm the exogeneity of the political shock (*Choc_Reel*). No economic variable exhibits significant predictive power over the shock (all $p > 0.14$).

Regarding the economic variables, the tests reveal a system characterized by high endogeneity and multiple feedback loops. A significant bidirectional causality is observed between *Industrial Production* and the *Balance of Payments* (BOP), with $p = 0.017$ for the $BOP \rightarrow Production$ vector and $p = 0.027$ for the $Production \rightarrow BOP$ vector. This indicates a reciprocal reinforcement between the financial external position and real output.

Furthermore, the price level (CPI) acts as a central node in the transmission mechanism. It Granger-causes the BOP at the 1% level ($p = 0.002$) and exhibits a bidirectional relationship with Industrial Production ($p = 0.008$ for $CPI \rightarrow Production$ and $p = 0.022$ for $Production \rightarrow CPI$). Collectively, these results demonstrate that once the exogenous shock is transmitted a series of internal feedback mechanisms between inflation, external accounts, and real production are activated.

Table 4: Granger Causality Wald Tests (Quarterly VAR Model)

Equation	**Excluded Variable**	**chi2**	**df**	**Prob chi2**
Choc_Reel	std_d_cpi	3.556	4	0.469
	std_d_prod	1.683	4	0.794
	std_d_bop	6.843	4	0.144
std_d_cpi	Choc_Reel	9.105	4	0.059*
	std_d_prod	11.447	4	0.022**
	std_d_bop	5.611	4	0.230
std_d_prod	Choc_Reel	1.484	4	0.829
	std_d_cpi	13.717	4	0.008***
	std_d_bop	11.989	4	0.017**
std_d_bop	Choc_Reel	0.888	4	0.926
	std_d_cpi	16.519	4	0.002***
	std_d_prod	10.981	4	0.027**

Note: *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. All variables are standardized and first-differenced.

6.5 Synthesis: Annual vs. Quarterly Findings

Comparing these results, a clear pattern emerges: the 2016 shock provoked a shift in institutional and social variables (Civil Liberties, Prison Rates) while causing high-frequency volatility in economic variables (BOP, Production) that is only clearly identified in the quarterly model.

7 Conclusion

This research investigated the multidimensional impact of the 2016 attempted coup on Turkey's institutional, economic, and social landscape. By employing a dual Vector Autoregression (VAR) framework, I have identified the specific channels through which political instability propagates into the real economy and the social fabric.

7.1 Summary of Empirical Findings

The empirical results provide evidence of a break. The annual model reveals that the 2016 shock triggered a permanent contraction in *civil liberties* and *freedom of expression*, which was statistically significant and persistent. Parallelly, the surge in the *prison rate* unaccompanied by a corresponding rise in *crime rates* confirms that the shock's primary social manifestation was a state-led judicial expansion rather than an increase in public insecurity.

The quarterly analysis reveals an interdependency the Balance of Payments and Industrial Production creating a cycle of instability.

7.2 Theoretical Implications: The Coup Trap and Social Decay

The findings align with the "coup trap" theory, where institutional instability creates a feedback loop that undermines economic growth. Furthermore, the observed decay in *Hope Levels* post-2019 suggests that while the population initially exhibited a "rally-around-the-flag" resilience, the cumulative effect of democratic backsliding and moral exclusion has led to a long-term erosion of social capital. The "social death" associated with emergency decree-laws represents a significant loss of human capital that likely acts as a drag on Turkey's long-term productivity.

8 Policy Implication

Based on these results, two primary policy implications emerge:

- **Institutional Decoupling:** Turkey must prioritize the independence of its judiciary and media to decouple political events from institutional quality. This is essential to restoring investor confidence and stabilizing FDI inflows.
- **Social Reintegration:** Addressing the exclusion of KHK victims and restoring moral boundaries is not only a human rights imperative but also an economic one to mitigate the rising "social death" and its associated fiscal and psychological costs.

In conclusion, the 2016 shock serves as a stark reminder that in an interconnected global economy, institutional decay is not merely a political concern but a fundamental to macroeconomic stability and social cohesion.

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9 Appendix

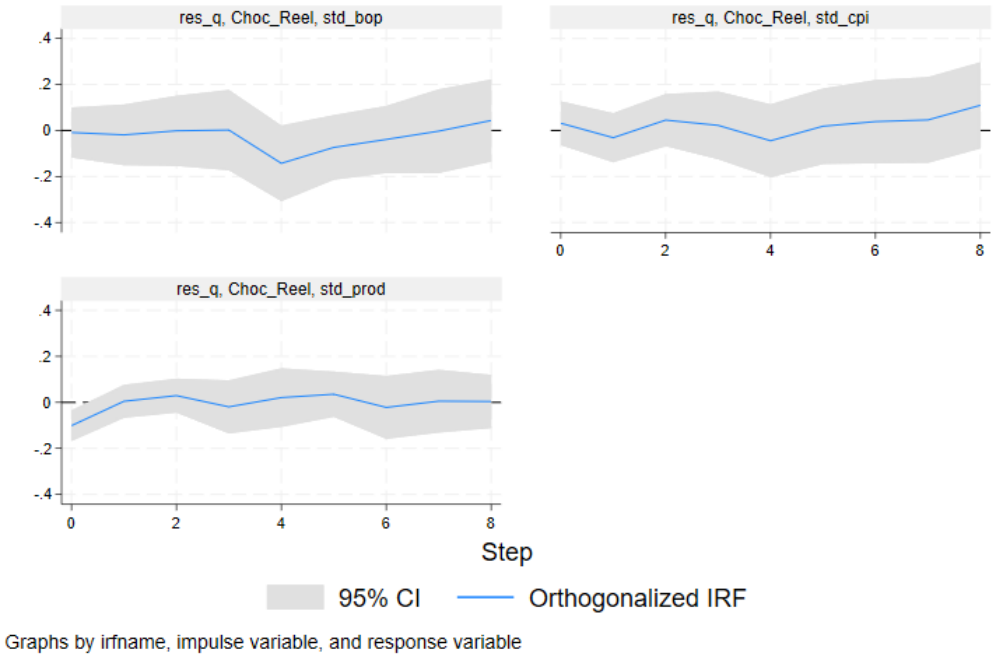

Figure 4: Quarterly IRF Standardized for BOP, CPI and Production

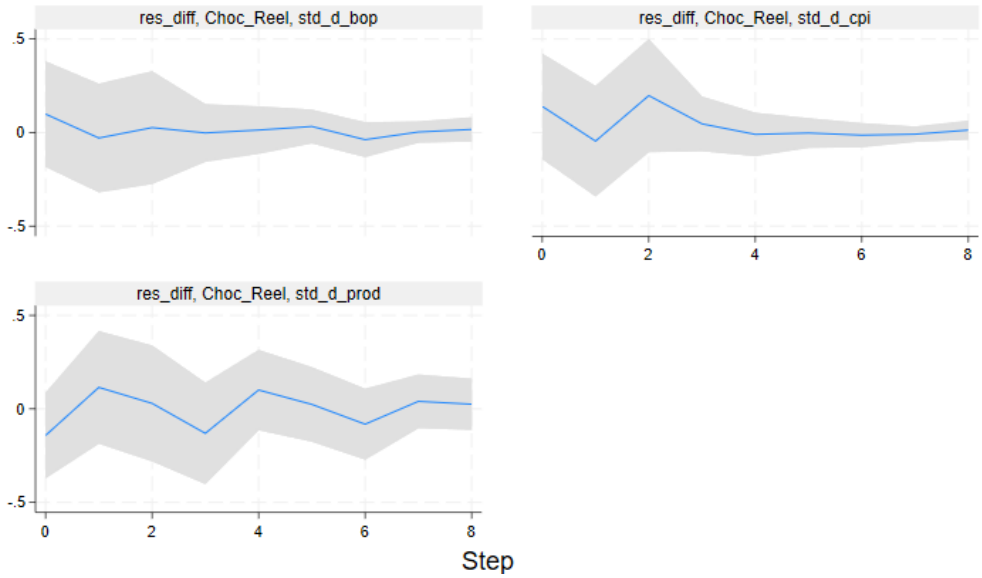

Figure 5: Quarterly IRF: Transmission from a Balance of Payments Shock to Industrial Production and CPI

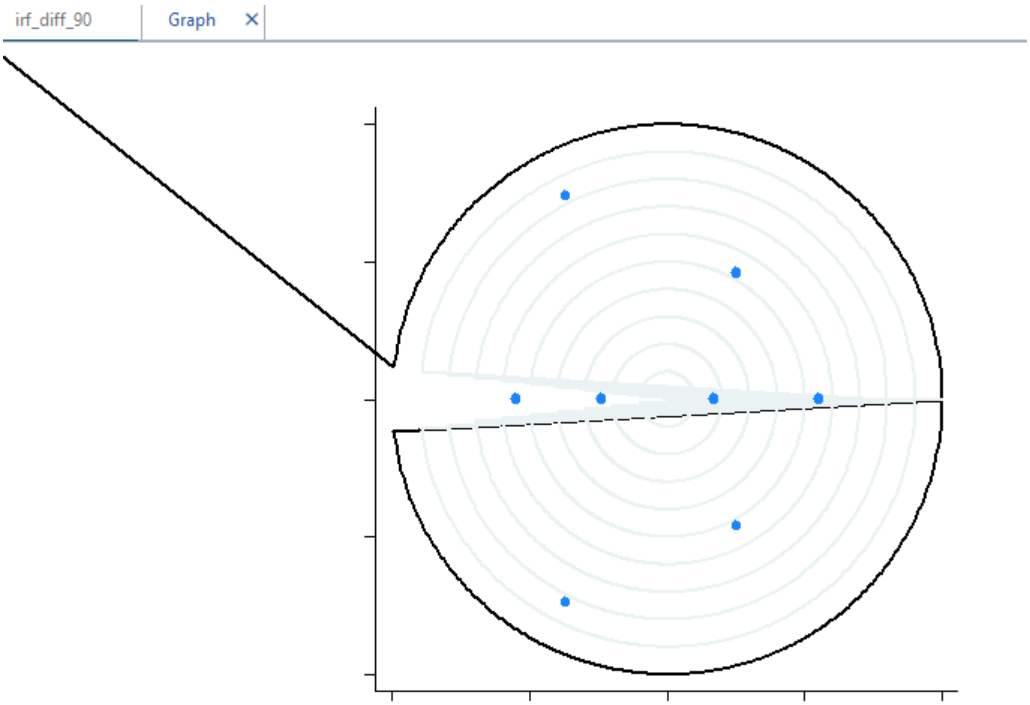

Figure 6: VAR Stability

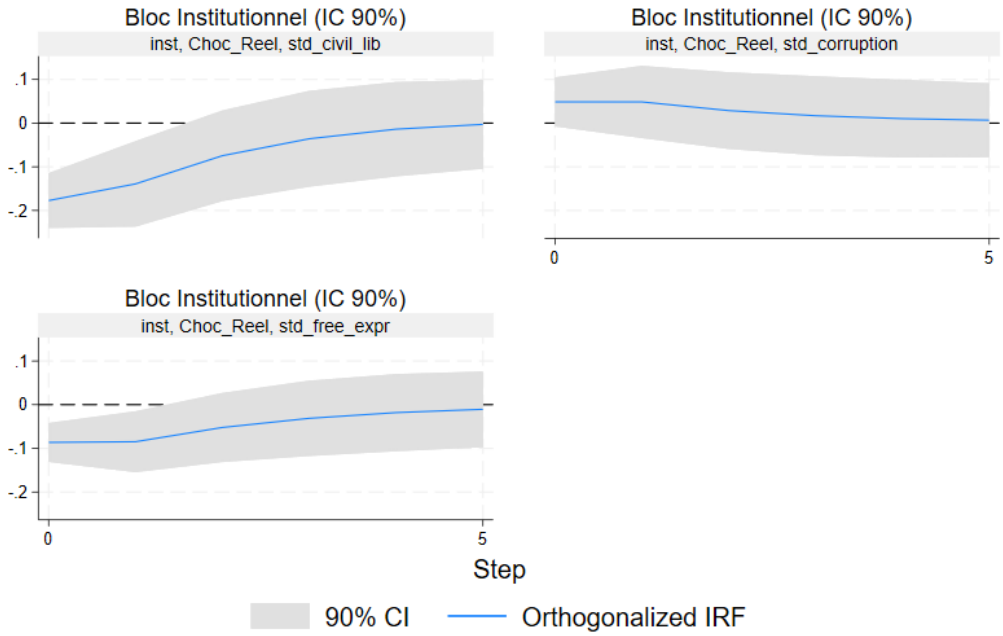

Figure 7: IRF Civil liberties, Corruption and free speach

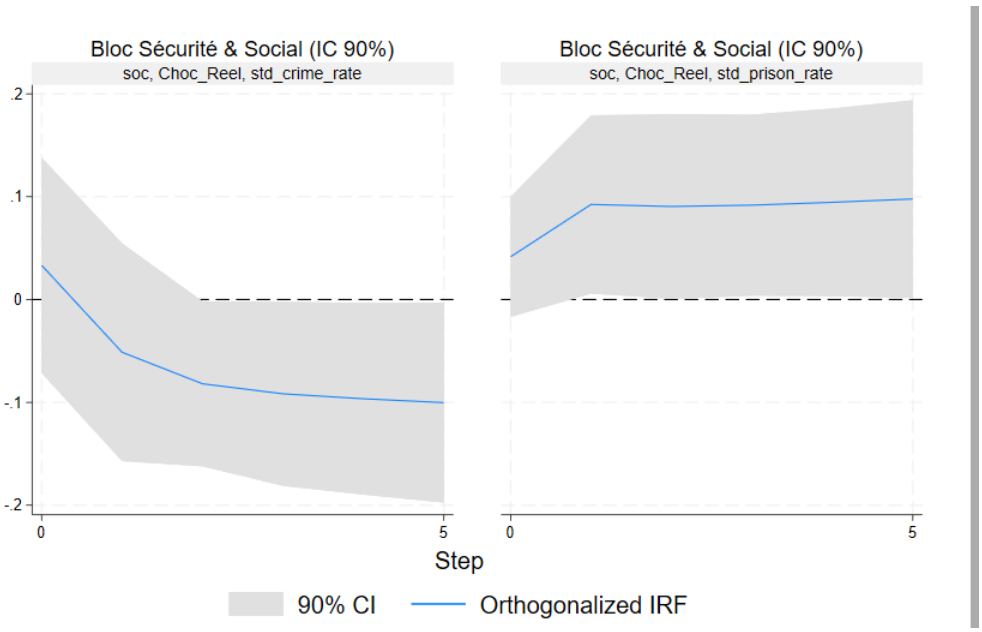

Figure 8: IRF Crime rate per 100k population and number of prisoner per 100K people

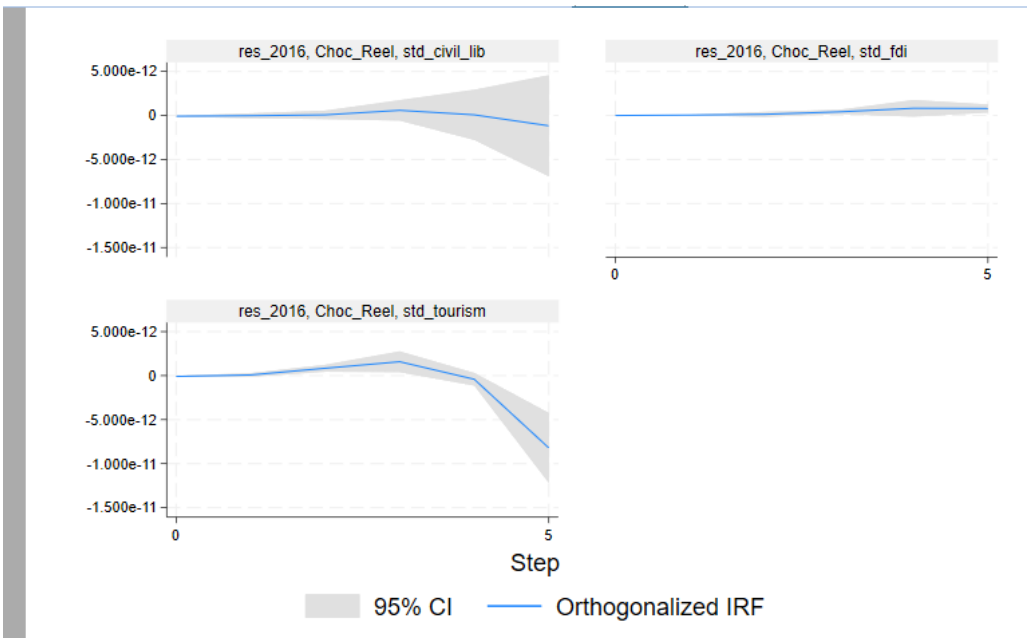

Figure 9: Standardized IRF of Civil Liberties, FDI, and Tourism

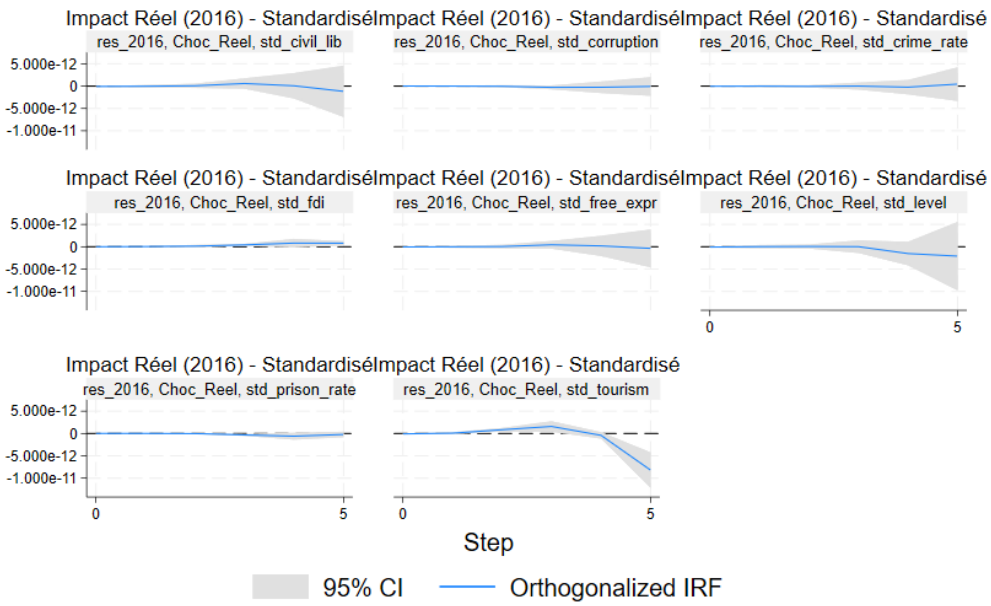

Figure 10: Standardized IRF of All annual Variables

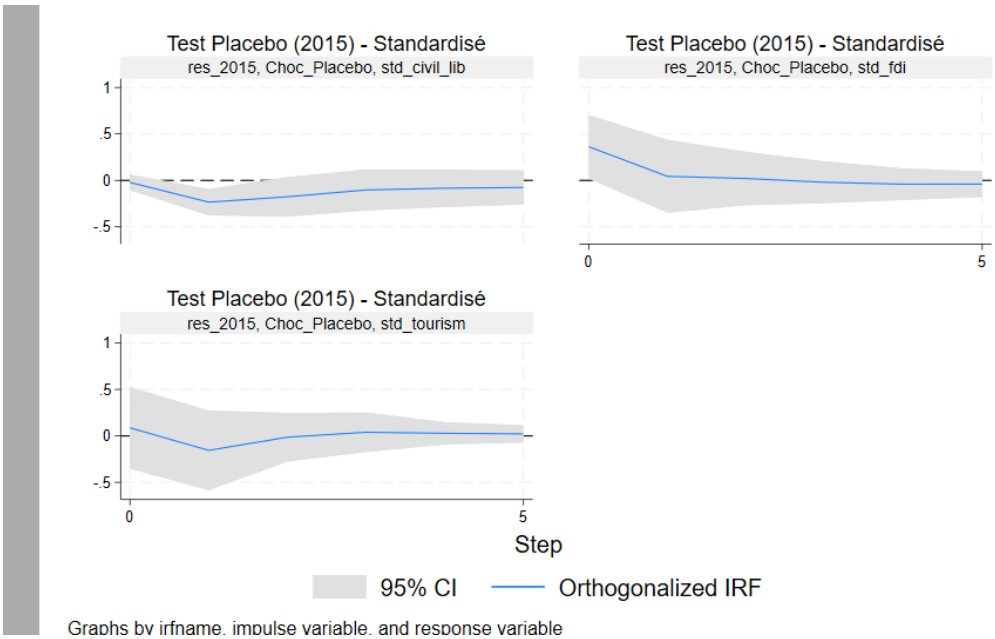

Figure 11: Placebo Test Civil liberties, FDI and Tourism revenue

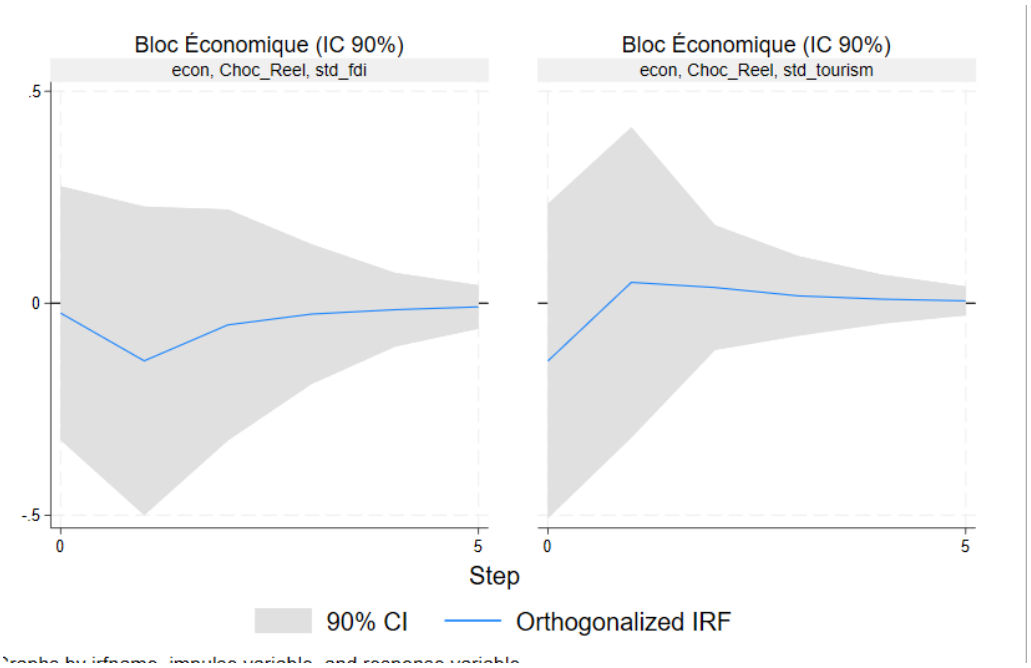

Figure 12: IRF FDI and Tourism revenue